

Ivan Evdokimov, Ph.D.

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Professional Summary

Software Engineer with 4+ years deploying enterprise-scale ML and GenAI solutions into production. Expert in building end-to-end MLOps pipelines with model governance, drift detection, and automated retraining on AWS. Proven track record of working in diverse multi-national teams and taking AI models from experimentation to production, processing sensitive social sciences datasets with 95%+ accuracy while ensuring scalable deployment.

Experience

Software Engineer – GenAI, University of Essex — Colchester, UK Nov 2025 — Present

- Designed and implemented GitHub Actions and Vercel CI/CD pipelines for an AI-assisted development workflow, enabling automated testing, validation, and weekly release cycles across a multi-team internal platform.
- Built end-to-end MLOps infrastructure for an internal platform serving multiple engineering and research teams, accelerating feature delivery from monthly to weekly releases through automated deployment and monitoring pipelines.
- Full-stack developed a native mobile application for an engineering client, enabling 20 field technicians to submit inspection reports via photos, recordings, and text; integrated the OpenAI API to automatically generate well-formed reports written directly to a database, saving 3 weeks of manual data collection annually and improving client communication."

Software Engineer – GenAI, UK Data Service — Colchester, UK Jan 2023 — Oct 2025

- Trained, fine-tuned, and deployed domain-specific PyTorch and scikit-learn models using Hugging Face, vLLM, and Ollama for automated metadata classification and disclosure risk assessment on a public data portal, achieving 95%+ accuracy on sensitive social science datasets.
- Designed and operated AWS-based ML retraining pipelines (Lambda, Step Functions, EFS) with drift and accuracy monitoring to trigger automated retrains, reducing manual intervention by 50% and maintaining 95%+ model accuracy on sensitive datasets.
- Built ETL pipelines to extract, transform, and query PostgreSQL-hosted social science datasets as model inputs for automated metadata classification, improving throughput and accuracy across large-scale sensitive records.
- Optimized statistical disclosure control algorithms by migrating from R to C++ using bitmask techniques, achieving 11x performance improvement and enabling real-time processing of datasets with 500k+ records.
- Contributed React components to a large-scale public data portal, building search, metadata filtering, and dataset preview features used by researchers and academics to browse and download sensitive social science datasets.
- Collaborated with academics, project managers, and public sector clients through regular requirements meetings, translating diverse non-technical needs into production ML system specifications.

Research Officer & ML Instructor, University of Essex — Colchester, UK Oct 2021 — Dec 2022

- Migrated quantitative macroeconomic models from MATLAB to C++, reducing simulation runtime 8x for collaborations with other universities in the UK and the USA.
- Taught Data Structures, C/C++, and ML to 100+ undergrad/postgrad students.

Projects

Chat-LLM github.com/ivan020/ChatLLM

- Built a self-hosted Ollama LLM server on Raspberry Pi 5, integrating with Twitch chat to enable viewers to submit prompts and receive live AI-generated responses during streams.

FinBot: Chat-Based Trading Simulation github.com/ivan020/FinBot

- Built a real-time portfolio tracker in Go with buy/sell mechanics, a PostgreSQL (Supabase) backend recording all transactions, and a live leaderboard — using Twitch chat as the user interface.
- Developed a React frontend with real-time portfolio visualisation, interactive performance graphs, and transaction history, serving 120+ monthly users.

MTG Card Reader ivan020.github.io/MtgRecognizer/

- Built a Golang REST API that uses OCR to extract text from Magic: The Gathering card photos, identifies the card via LLM, and returns real-time price data from a PostgreSQL-backed catalogue.
- Developed a React frontend with card filtering, deck management, and real-time price display, deployed on Vercel.

Education

University of Essex, PhD in Computational Finance Oct 2021 — May 2025

- Developed PyTorch-based transfer learning framework for Bayesian model averaging in financial forecasting; published peer-reviewed papers and released open-source package adopted externally.
- First-authored a peer-reviewed publication in a CS/ML journal on Bayesian model averaging for financial forecasting, and co-authored 3 international conference papers.

- Built C++ agent-based financial simulation modeling macroeconomic dynamics under negative interest rates.